

Decoder search and the support-recovery bottleneck

BP/EP, MAP-ADMM, NNOMP, VAMP, block MAP, and the decisive oracle experiment

Stage	2 of 5
Updated	5 September 2026
Status	Jobs 001–017 complete; algorithms, oracle knowledge, and conclusions separated

This report is a chronological research record. Measured results, interpretations, and open hypotheses are labelled separately. Detailed numerical ledgers remain in **results/**.

1 Why every decoder was put in one model

The initial work mixed a structured ODMA codebook with a specialised approximation. A poor result could therefore be caused by the physical columns, the support search, the prior, the numerical iteration, or channel uncertainty. The next stage fixed one global target for every method:

$$y = \Phi a + z, \text{quad}\Phi = [\phi_0, \dots, \phi_{M-1}] \in \mathbb{C}^{n \times M}, \text{quad}M = 2^B,$$

$$a \in \mathbb{Z}_+^M, \text{quad}\mathbf{1}^T a = K_a, \text{quad}z \sim \mathcal{CN}(0, \sigma_{\text{eff}}^2 I_n), \text{quad}\|\phi_m\|_2 = 1.$$

Dense, sparse-global, and ODMA constructions now differed only through Φ . Every receiver returned the same count vector, allowing the experiment to separate codebook geometry from decoder behaviour.

For the common-signature multi-antenna model of Report 1,

$$\bar{y} = Y h^* / \|h\|_2^2, \text{quad}\sigma_{\text{eff}}^2 = \sigma^2 / \|h\|_2^2,$$

so the remaining algebra is scalar without discarding signal information under that model.

2 Exact likelihood, prior, and recovery target

Conditioned on a ,

$$-\log p(y | a) = \frac{1}{\sigma_{\text{eff}}^2} \|y - \Phi a\|_2^2 + \text{const.}$$

If exactly K_a devices independently choose one of M messages uniformly, the count vector is multinomial:

$$p(a | K_a) = \frac{K_a!}{M^{K_a} \prod_{m=1}^M a_m!} \mathbf{1}\{a \in \mathbb{Z}_+^M, \mathbf{1}^T a = K_a\}.$$

Hence exact MAP with known K_a is

$$\hat{a}_{\text{MAP}} = \arg \min_{a \in \mathbb{Z}_+^M, \mathbf{1}^T a = K_a} \frac{1}{\sigma_{\text{eff}}^2} \|y - \Phi a\|_2^2 + \sum_{m=1}^M \log(a_m!).$$

The pure maximum-likelihood reference omits the factorial term. Both are combinatorial: there are

$$\binom{M + K_a - 1}{K_a}$$

nonnegative integer count vectors of total K_a . Exact enumeration is therefore a tiny-system certificate.

When K_a is unknown, the research code also used a softened total-count prior. Let $K = \mathbf{1}^T a$ and $K \sim \mathcal{N}(\mu_K, \sigma_K^2)$. Up to constants, the variable- K multinomial objective is

$$J(a) = \frac{\beta}{2} \|y - \Phi a\|_2^2 + \sum_m \log(a_m!) - \log(K!) + K \log M + \frac{(K - \mu_K)^2}{2\sigma_K^2}, \quad \beta = 2/\sigma_{\text{eff}}^2$$

under the implementation's real-Gaussian scaling. This objective is the common reference for the ADMM and block coordinate families below.

3 Decoder family 1: Graph-BP/EP

The Graph-BP algorithm in Report 1 works in ODMA blocks, approximates cross-block interference by Gaussians, then enumerates each local count state. In the global notation it approximates the exact posterior

$$p(a | y) \propto \exp[-\|y - \Phi a\|_2^2 / \sigma_{\text{eff}}^2] p(a)$$

by block Gaussian sites. Its attraction is structural locality; its weaknesses are loopy overconfidence, a restricted local state cache, and an independent-Poisson approximation to the global multinomial dependence. It remains a useful historical structured receiver, but the later global experiments did not rely on it for the decisive comparison.

4 Decoder family 2: MAP-ADMM

4.1 Variable split

ADMM separates a continuous data-fit variable $x \in \mathbb{R}_+^M$ from a structured count variable a :

$$\min_{x,a} f(x) + g(a) \quad \text{s.t.} \quad x = a,$$

$$f(x) = \frac{\beta}{2} \|y - \Phi x\|_2^2 + I_{\mathbb{R}_+^M}(x),$$

where g is either the independent-Poisson penalty, the multinomial/soft- K penalty, or an indicator restricting a to cached block states. With scaled dual u and penalty $\rho > 0$,

$$\mathcal{L}_\rho(x, a, u) = f(x) + g(a) + \frac{\rho}{2} \|x - a + u\|_2^2 - \frac{\rho}{2} \|u\|_2^2.$$

One iteration is

$$\begin{aligned} x^{t+1} &= \arg \min_{x \geq 0} f(x) + \frac{\rho_t}{2} \|x - a^t + u^t\|_2^2, \\ a^{t+1} &= \arg \min_a g(a) + \frac{\rho_t}{2} \|x^{t+1} - a + u^t\|_2^2, \\ u^{t+1} &= u^t + x^{t+1} - a^{t+1}. \end{aligned}$$

4.2 Continuous count-relaxation update

Ignoring the nonnegative boundary, stationarity gives

$$(\beta \Phi^H \Phi + \rho I)x = \beta \Phi^H y + \rho(a^t - u^t).$$

Thus

$$x_{\text{unc}}^{t+1} = (\beta \Phi^H \Phi + \rho I)^{-1} [\beta \Phi^H y + \rho(a^t - u^t)].$$

Projection $[x_{\text{unc}}]_+$ is exact only when the quadratic metric is diagonal. The implementation therefore uses iterative nonnegative solution steps; this distinction matters when interpreting it as MAP rather than a heuristic. Using Woodbury can move the dense inverse to resource dimension:

$$(\rho I + \beta \Phi^H \Phi)^{-1} = \rho^{-1} I - \rho^{-1} \Phi^H (\beta^{-1} I + \rho^{-1} \Phi \Phi^H)^{-1} \Phi \rho^{-1}.$$

4.3 Independent-Poisson discrete update

For $a_m \sim \text{Poisson}(\lambda)$ independently,

$$g(a) = \sum_m [\log \Gamma(a_m + 1) - a_m \log \lambda] + I_{\mathbb{Z}_+^M}(a) + \text{const.}$$

Writing $v = x^{t+1} + u^t$, the update separates:

$$a_m^{t+1} = \arg \min_{c \in \mathbb{Z}_+} \log \Gamma(c + 1) - c \log \lambda + \frac{\rho}{2}(c - v_m)^2.$$

Only a small range of c needs evaluation at low collision. An EM-style rate update is $\lambda^+ = M^{-1} \sum_m a_m$.

4.4 Multinomial and fixed/soft load updates

The terms $-\log(K!) + K \log M + (K - \mu_K)^2 / (2\sigma_K^2)$ couple all coordinates through K . The implementation groups variables into ODMA blocks and enumerates a finite state set $\mathcal{A}_q$ for each. For state $s \in \mathcal{A}_q$, record its local count $k_q(s) = \mathbf{1}^T s$ and local cost. A dynamic programme computes

$$D_q(k) = \min_{\substack{s_1 \in \mathcal{A}_1, \dots, s_q \in \mathcal{A}_q \\ \sum_{j=1}^q k_j(s_j) = k}} \sum_{j=1}^q J_j(s_j).$$

Oracle- K chooses $D_Q(K_a)$. Soft- K chooses

$$K^* = \arg \min_k D_Q(k) - \log(k!) + k \log M + \frac{(k - \mu_K)^2}{2\sigma_K^2}.$$

Backtracking returns one state per block. This enforces the cached-domain total-count logic exactly, but the cache itself may exclude the truth.

4.5 Penalty and acceleration variants

Residual balancing, spectral- ρ , SRA- ρ , and Anderson acceleration all modify optimisation dynamics. They do not change the statistical objective or remove the restricted-state assumption. Primal and dual residuals are

$$r^{t+1} = x^{t+1} - a^{t+1}, \quad s^{t+1} = \rho_t(a^{t+1} - a^t).$$

Residual balancing increases ρ if $\|r\|$ dominates and decreases it if $\|s\|$ dominates. Spectral and SRA variants estimate local curvature from successive residual/dual differences. Anderson extrapolates the fixed-point map. These variants were retained as diagnostics rather than evidence of a different decoder class.

5 Decoder family 3: NNOMP

NNOMP uses the same global dictionary but treats the count vector as a sparse nonnegative coefficient vector. Starting with $S_0 = \emptyset$ and $r^0 = y$, iteration t selects

$$j_t = \arg \max_{j \notin S_t} \Re\{\phi_j^H r^t\}, \quad S_{t+1} = S_t \cup \{j_t\}.$$

It then refits all selected coefficients:

$$x_{S_{t+1}} = \arg \min_{x \geq 0} \|y - \Phi_{S_{t+1}} x\|_2^2, \quad r^{t+1} = y - \Phi_{S_{t+1}} x_{S_{t+1}}.$$

The count output is obtained by nonnegative integer projection, normally with total K_a . NNOMP-OracleK stops after the known mass/support budget has been met. It is therefore a clear support-search baseline, but it is not a non-oracle or Bayes-optimal URA receiver. Greedy selection also makes it sensitive to correlation tails and to several active columns explaining the same residual direction.

6 Decoder family 4: VAMP

VAMP alternates a separable denoiser and a linear Gaussian estimator. For pseudo-observation

$$r_1 = a + \mathcal{N}(0, \gamma_1^{-1} I),$$

the Bernoulli-Gaussian implementation used

$$p(a_m) = (1 - \rho)\delta_0 + \rho\mathcal{N}(0, \sigma_x^2)$$

or a nonnegative discrete Poisson denoiser. The denoiser returns $x_1 = g_1(r_1, \gamma_1)$ and average sensitivity

$$\alpha_1 = M^{-1} \text{tr} \left(\frac{\partial g_1}{\partial r_1} \right).$$

Extrinsic conversion gives

$$\gamma_2 = \gamma_1 \frac{\alpha_1}{1 - \alpha_1}, \quad r_2 = \frac{x_1 - \alpha_1 r_1}{1 - \alpha_1}.$$

The linear module is

$$x_2 = \left(\sigma_{\text{eff}}^{-2} \Phi^H \Phi + \gamma_2 I \right)^{-1} \left(\sigma_{\text{eff}}^{-2} \Phi^H y + \gamma_2 r_2 \right).$$

Its average sensitivity α_2 produces

$$\gamma_1^+ = \gamma_2 \frac{\alpha_2}{1 - \alpha_2}, \quad r_1^+ = \frac{x_2 - \alpha_2 r_2}{1 - \alpha_2}.$$

SVD makes the linear update practical at the tested scale. EM variants update ρ , σ_x^2 , Poisson λ , and noise variance. The algorithm is best justified for suitable large random matrix ensembles; the ODMA dictionary does not automatically satisfy those assumptions, and this implementation is VAMP-like rather than a paper-specific state-evolution claim.

7 Decoder family 5: block MAP, block coordinate MAP, and SIC

Independent BlockMAP projects the matched observation onto each ODMA support and chooses

$$\hat{a}_q = \arg \min_{a \in \mathcal{A}_q} \frac{\beta}{2} \|P_q^H \bar{y} - C_q^T a\|_2^2 - \log p(a).$$

It ignores interference from other blocks and is therefore a local diagnostic.

Block coordinate MAP instead maintains $\hat{y} = \sum_q P_q C_q^T a_q$. Removing block q gives

$$r^{(-q)} = \bar{y} - \hat{y} + P_q C_q^T a_q.$$

For candidate $s \in \mathcal{A}_q$, $K_{-q} = \sum_{j \neq q} \mathbf{1}^T a_j$, and $k_q(s) = \mathbf{1}^T s$, its score is

$$J_q(s) = \frac{\beta}{2} \|P_q^H r^{(-q)} - C_q^T s\|_2^2 + \text{sum}_v \log(s_v!) \\ - \log((K_{-q} + k_q(s))!) + (K_{-q} + k_q(s)) \log M + \frac{(K_{-q} + k_q(s) - \mu_K)^2}{2\sigma_K^2}.$$

Choose $s_q^* = \arg \min_{s \in \mathcal{A}_q} J_q(s)$ and update

$$\hat{y} \leftarrow \hat{y} + P_q C_q^T (s_q^* - a_q).$$

The soft- K variant forms $q_q(s) \propto \exp[-J_q(s)/T]$ and averages the implied conditional total counts to update μ_K . This is pseudo-EM because the block distributions are conditional replacements, not independent marginals.

SIC uses the same residual principle but makes hard high-confidence decisions, subtracts their waveforms, and repeats. It can reduce interference after a correct decision, but error propagation makes its result strongly ordering-dependent.

7.1 Map from equations to retained implementation names

The decoder registry preserves the following distinctions. They are useful when reading old plots or logs:

- **Graph-BP**: the turbo-scheduled block EP algorithm with cavity-by-omission and Poisson/noise EM updates from Report 1.
- **ADMM-Poisson** / **ADMM-Multinom**: the common ADMM split with, respectively, independent-Poisson or softened multinomial load penalties.
- **ADMM-KDP-OracleK** / **ADMM-KDP-SoftK**: the same split with the block-state dynamic programme evaluated at known total load or selected under a soft load prior.
- **ADMM-KDP-SpectralRho** / **Anderson** / **SRA**: soft-load KDP with a different penalty adaptation or fixed-point acceleration; these change dynamics, not the statistical target.
- **BlockCD-OracleK** / **BlockCD-SoftK**: sequential discrete block replacement under a tight oracle load prior or a damped pseudo-EM load estimate.
- **VAMP-BG-OracleK**: Bernoulli-Gaussian VAMP whose activity moments are derived from the true load.
- **VAMP-BG-EMRho** / **EMRhoSigma** / **EMall**: successively learn activity probability, slab variance, and noise inside the VAMP approximation.
- **VAMP-Poisson-EM**: replaces the continuous active slab by a discrete nonnegative Poisson-count denoiser.
- **BlockMAP**: independent per-block exact search inside its cached state domain; it ignores cross-block interference.
- **NNOMP** / **NNOMP-OracleK**: greedy global support selection with a non-oracle stopping rule or the true load budget.
- **SIC**: global matched-filter cancellation with BIC stopping and no oracle load.

The common equations above show which differences alter the posterior approximation and which only alter optimiser behaviour. This prevents a long list of decoder labels from appearing to be a long list of independent physical models.

8 Metrics and collision convention

For true and estimated counts with equal total mass,

$$\text{PUPE}(a, \hat{a}) = 1 - \frac{1}{K_a} \sum_{m=1}^M \min(a_m, \hat{a}_m).$$

This becomes the missed-set fraction when counts are binary, while remaining correct when payload collisions occur. At small B , support-only evaluation is insufficient. Polyanskiy-style achievability curves use their own collision convention and are coding benchmarks, not expected performance curves for any decoder above.

9 Dense versus ODMA and the oracle-support control

Jobs 001–008 and 013–014 used matched (B, n, K_a) , 50 seeds, NNOMP-OracleK, and target PUPE ≤ 0.05 . Representative required E_b/N_0 values are:

Geometry and load	Dense	ODMA 1	ODMA 2	ODMA 3
$B = 12, n = 2048, K_a = 333$	1.00	3.69	3.25	> 4
$B = 12, n = 512, K_a = 52$	2.19	11.56	6.44	unbracketed
$B = 12, n = 256, K_a = 17$	2.19	9.12	5.75	5.56

Table 1: Required E_b/N_0 in dB. ODMA columns use different placement arrangements; the table demonstrates a practical algorithmic gap, not a universal ordering of every ODMA design.

The decisive experiment supplies the true support $S = \text{supp}(a)$ and solves only

$$\hat{a}_S = \arg \min_{x \geq 0} \|y - \Phi_S x\|_2^2,$$

followed by integer/total-mass projection. This removes support search but preserves noise, interference, active-column conditioning, and multiplicity estimation.

Geometry	Mean oracle ODMA-minus-dense gap	Non-oracle contrast
$B = 12, n = 2048$	0.05–0.11 dB	Practical gap reaches 2+ dB at high load
$B = 12, n = 512$	0.10, 0.12, 0.28 dB	NNOMP support loss is about 5–6 dB
$B = 12, n = 256$	0.08, −0.01, 0.03 dB	NNOMP support loss is about 6–7 dB

Table 2: Jobs 012, 016, and 017. The true support is an explicit oracle.

Evidence status: Measured result. In the tested low-collision $B = 12$ regimes, the active ODMA columns were nearly as easy as dense columns to fit once their identities were known. Most of the observed NNOMP gap arose while locating support.

This is not an information-theoretic proof that ODMA has no geometric penalty, nor evidence that an ideal ODMA decoder must close the gap. It is a causal result about the tested matrices and recovery pipeline. It redirected the work from adding increasingly elaborate inference updates to changing the searchable codebook family itself.

Report 3 therefore starts from the unrestricted explicit global matrix, derives a factorisation that contains dense, sparse-global, product, ODMA, and sectioned codebooks, and asks which restrictions preserve the good global landscape.